

# ITAI-2373

# Natural Language Processing in 2026

Language Understanding in the Era of Large Language Models

-Srikanth Bhakthan

# Welcome

- **Our Goal:** To move you from "using" ChatGPT to "building" the agentic AI systems that power it and understanding the mechanics underneath.
- **Logistics:**
  - Lecture: Monday 5-7pm CST
  - Lab: Do it at your own pace & time
  - Office Hours: Ask in Teams/ Canvas email and I will respond as and when I have time

# Course Assessment & Expectations

- **How you will be graded:**
  - **Lab Assignments (~35%):** Hands-on AI driven coding (Python/Jupyter notebook-Google Colab). You can use your favorite coding agent!
  - **Quiz(~35%):** Concepts and Theory.
  - **Final Project (~35%):** Building a working NLP/AI/Agent prototype.
- **Expectations:**
  - This is a fast-moving field. We will learn foundational theory, but we will also experiment with tools that were *just* released.
  - Active participation in "Red Teaming" (breaking models) and ethical discussions is required.

# The Big Picture

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# THE EVOLUTIONARY TIMELINE OF ARTIFICIAL INTELLIGENCE

(1940s – FUTURE SINGULARITY)

1940s:  
The Theoretical  
Roots



1940s:  
The Theoretical  
Roots

Early computing  
& cryptography.  
Logic gates and  
machine theory.

1950s-60s:  
The Mainframe Era



1950s-60s:  
The Mainframe  
Era

Bulk processing  
& data storage.  
First AI programs.

1970s-80s:  
The Symbolic Age



1970s-80s:  
The Symbolic  
Age

Expert Systems  
& rule-based AI.  
Knowledge  
representation.

1990s:  
The Connectionist  
Web



1990s:  
The Connectionist  
Web

Rise of the internet  
& neural networks.  
Data connectivity.

2010s:  
The Deep Learning  
Boom



2010s:  
The Deep Learning  
Boom

Big Data & GPU  
acceleration.  
Multi-layered  
neural nets.

2020s:  
The Generative  
Transformers



2020s:  
The Generative  
Transformers

Large Language  
Models &  
generative AI.  
Content creation.

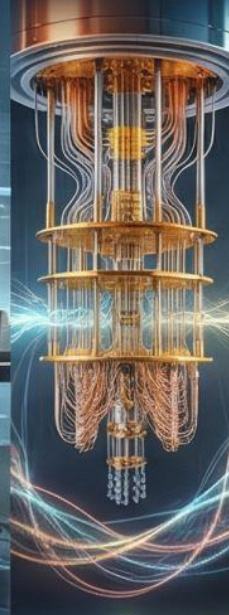
Near Future:  
The Agentic  
Integration



Near Future:  
The Agentic  
Integration

AI in physical  
world. Robotics &  
autonomous  
agents.

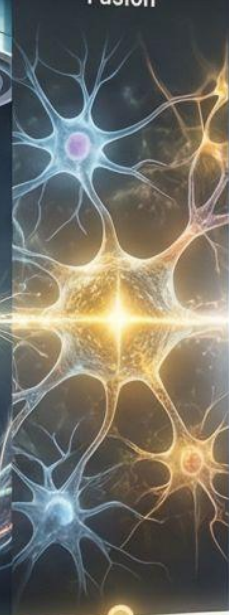
Future:  
The Quantum Leap



Future:  
The Quantum  
Leap

Exponential  
processing speed.  
Quantum  
algorithms.

Far Future:  
The Bio-Digital  
Fusion



Far Future:  
The Bio-Digital  
Fusion

Merging biology  
& machine.  
Augmented  
intelligence.

Final Stage:  
The AGI  
Singularity



Final Stage:  
The AGI  
Singularity

Infinite intelligence  
& self-improvement.  
Beyond human  
comprehension.

VISUAL EVOLUTIONARY PROGRESSION OF ARTIFICIAL GENERAL INTELLIGENCE (AGI)

# The Robotics Connection

*Why does a Robotics major need NLP? (B.A.T)*

## **1. Human-Robot Interaction (HRI):**

"Robot, go to the kitchen and fetch the red mug." (Voice to Action).

## **2. Knowledge Retrieval:**

Robots reading manuals to learn how to fix machinery.

## **3. Multimodality:**

Modern AI sees, hears, and speaks.

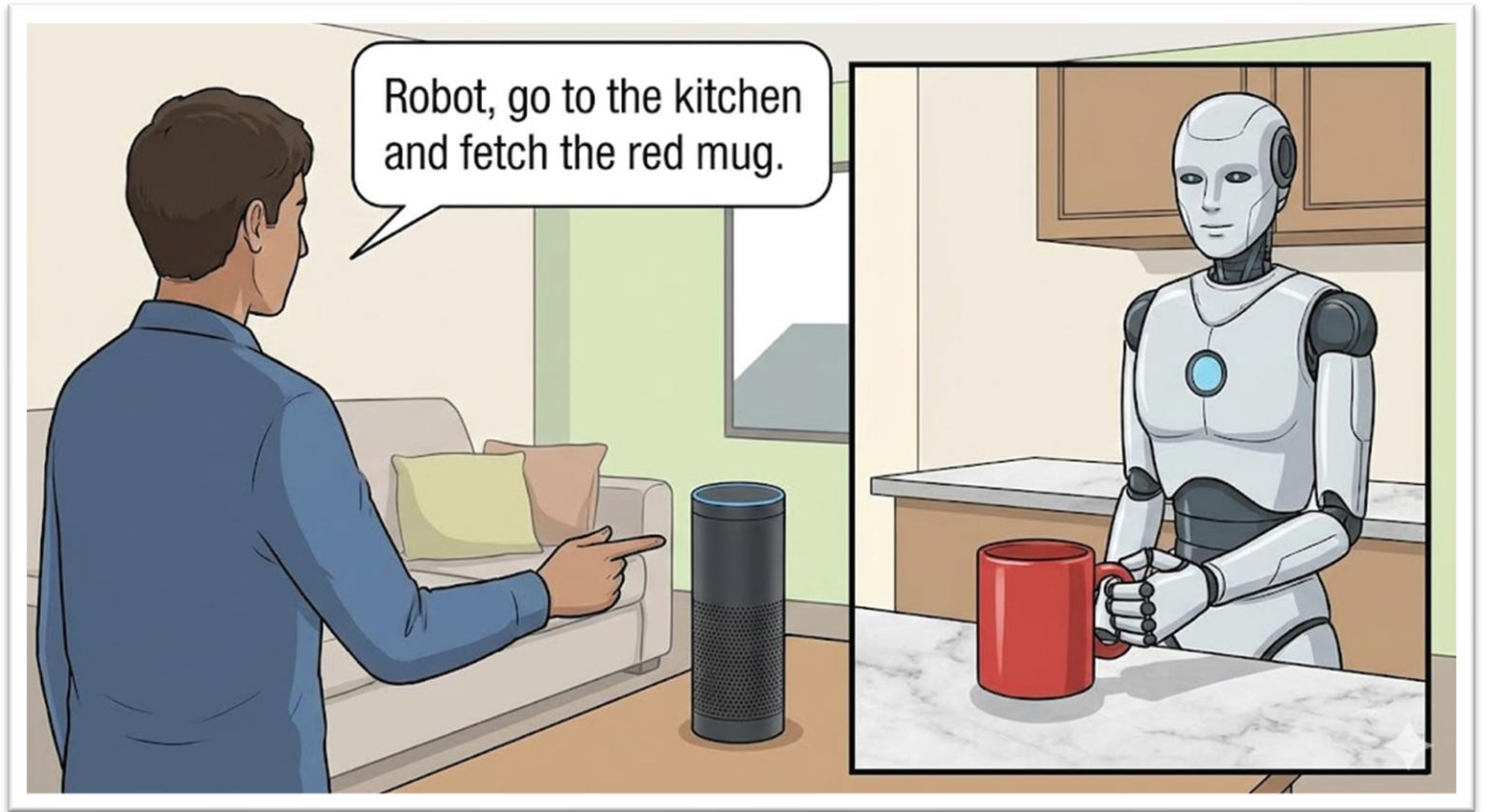
NLP is the bridge between these modalities.



# 1. Human-Robot Interaction (HRI)

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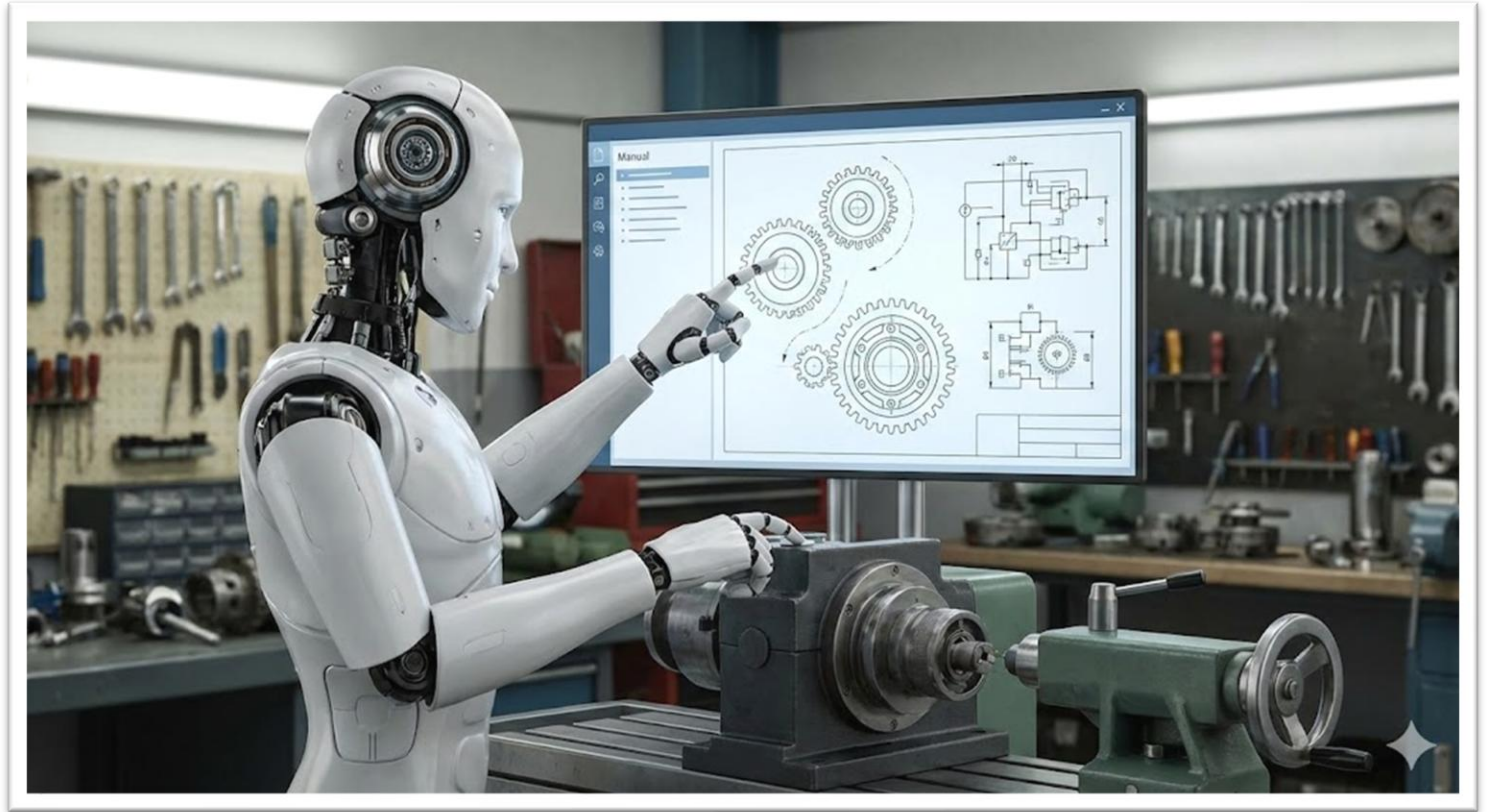
- The integration of LLMs in robotics allows for more natural and intuitive human-robot interaction.
- A user can give a complex, natural language command, and the robot, powered by an LLM,
- can interpret the intent and translate it into a sequence of actions.



## 2. Knowledge Retrieval

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- Robots often need to perform tasks in dynamic environments or on complex machinery.
- LLMs can act as a vast knowledge base, allowing robots to "read" manuals, technical documents, or online tutorials to learn how to fix a machine or perform a new task, without being explicitly programmed for it.



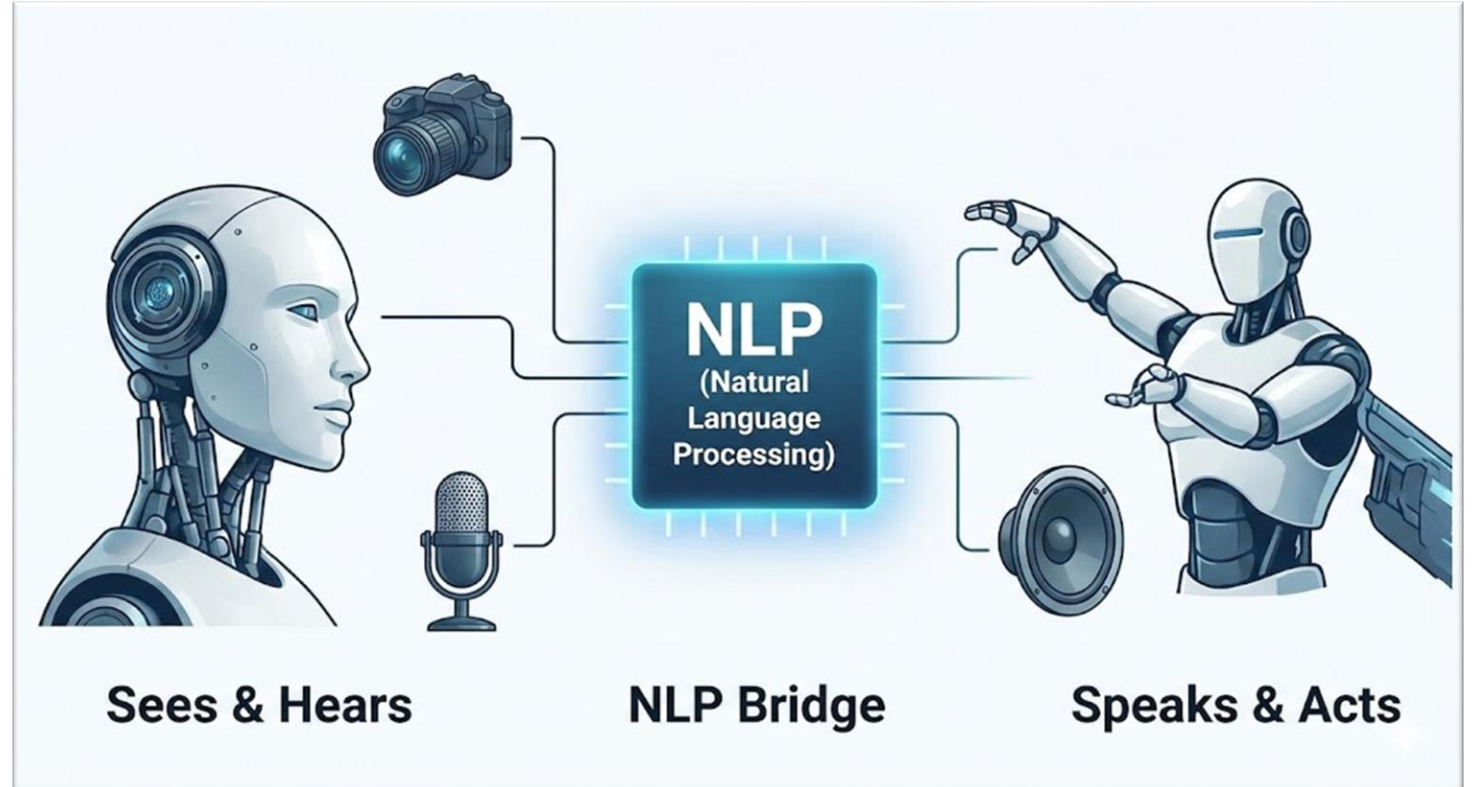


### 3.

# Multimodality

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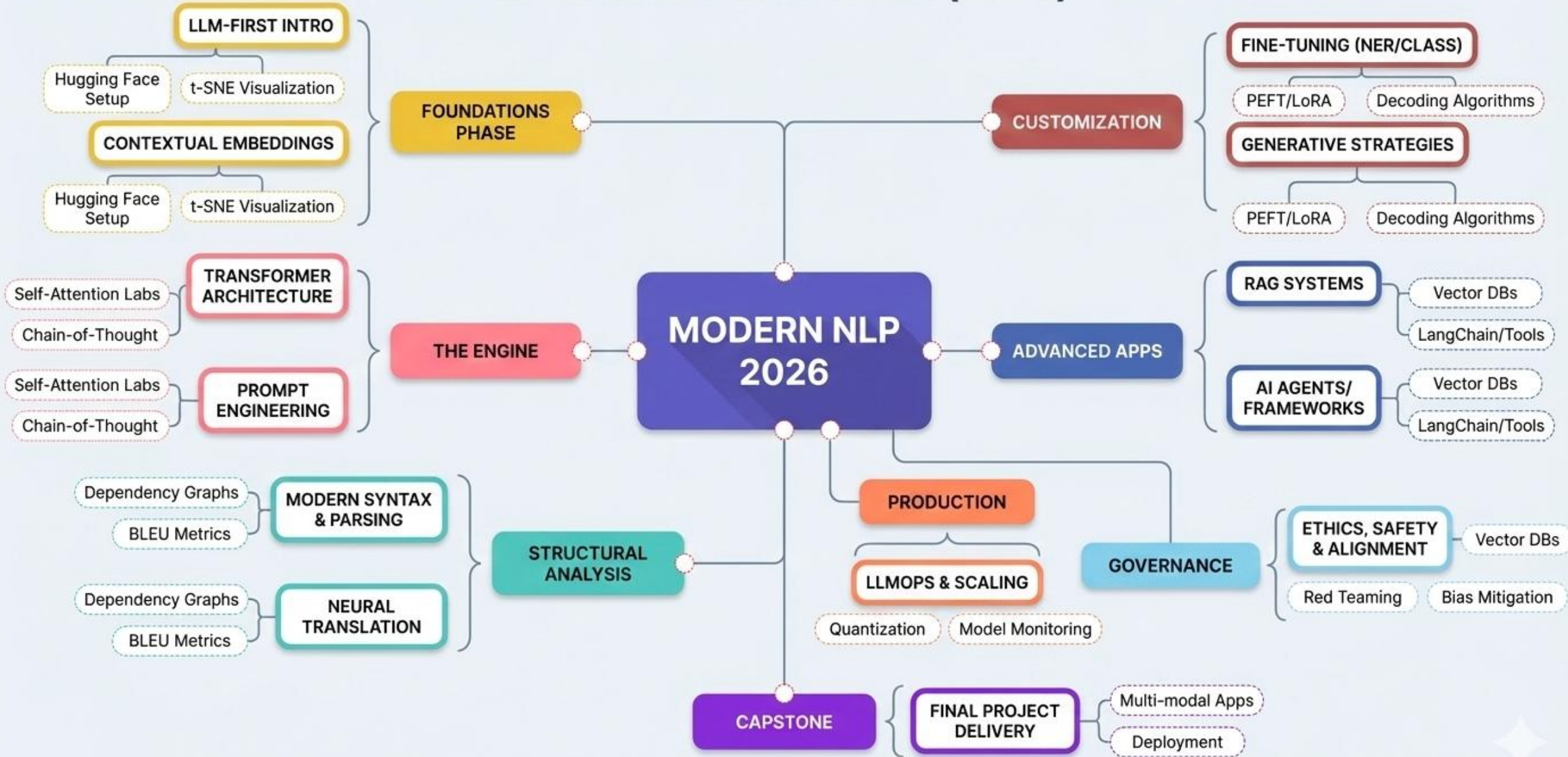
- Modern Robotics is moving towards multimodal AI, where a robot can perceive the world through various senses like vision and hearing.
- Then use NLP as a central “brain” to process this information and decide on the appropriate action, including speaking back to human.





# The Paradigm Shift

# Modern NLP Curriculum (2026)





# Modern NLP Curriculum (2026): An LLM-First Paradigm Shift

## Part 1: Foundations of Modern NLP



**Module 1:** Introduction to the LLM-First World



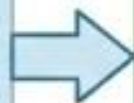
**Module 2:** Text Representation: The Power of Embeddings



**Module 3:** The Engine of Modern NLP: The Transformer Architecture



**Module 4:** Prompt Engineering & In-Context Learning



## Part 2: Customizing and Applying Language Models



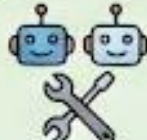
**Module 5:** Fine-Tuning Pre-Trained Models (NER & Classification Focus)



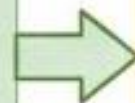
**Module 6:** Advanced Text Generation & Summarization



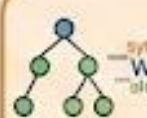
**Module 7:** Retrieval-Augmented Generation (RAG)



**Module 8:** Building LLM-Powered Chatbots & Agents



## Part 3: The Broader Ecosystem and Societal Impact



**Module 9 & 10:** Syntax, Parsing, and Topic Modeling (A Modern View)



**Module 11:** Machine Translation in the Transformer Era



**Module 12:** The LLM Ecosystem & MLOps



**Module 13 & 14:** Ethics, Safety, and Alignment



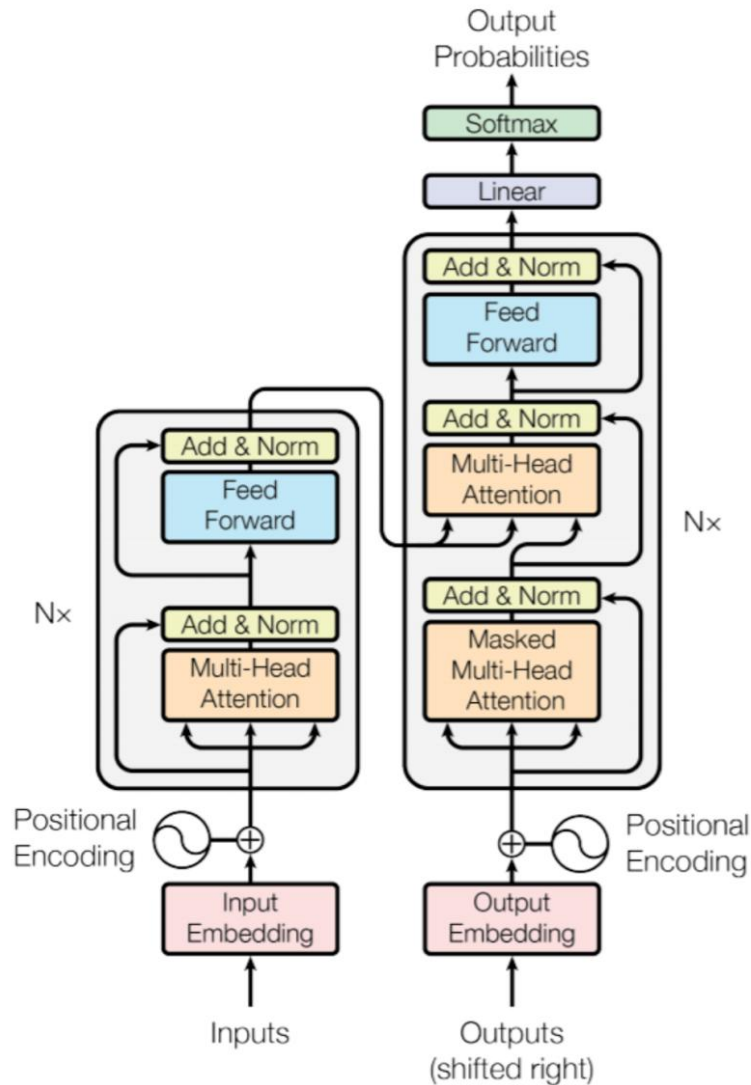
**Module 15 & 16:** Final Project & Course Wrap-Up



# What is Natural Language Processing?

- **Definition:** A subfield of AI giving computers the ability to understand, interpret, and generate human language.
- **The Challenge:** Human language is ambiguous, messy, and context-dependent.
  - *Example:* "I saw the man with the telescope." (Did I have the telescope? Or did the man?)
  - *Example:* "Bank" (River? Money? Turning a plane?)

# The "Two Eras" of NLP



- **Era 1: Pre-Transformer (1950s - 2017)**

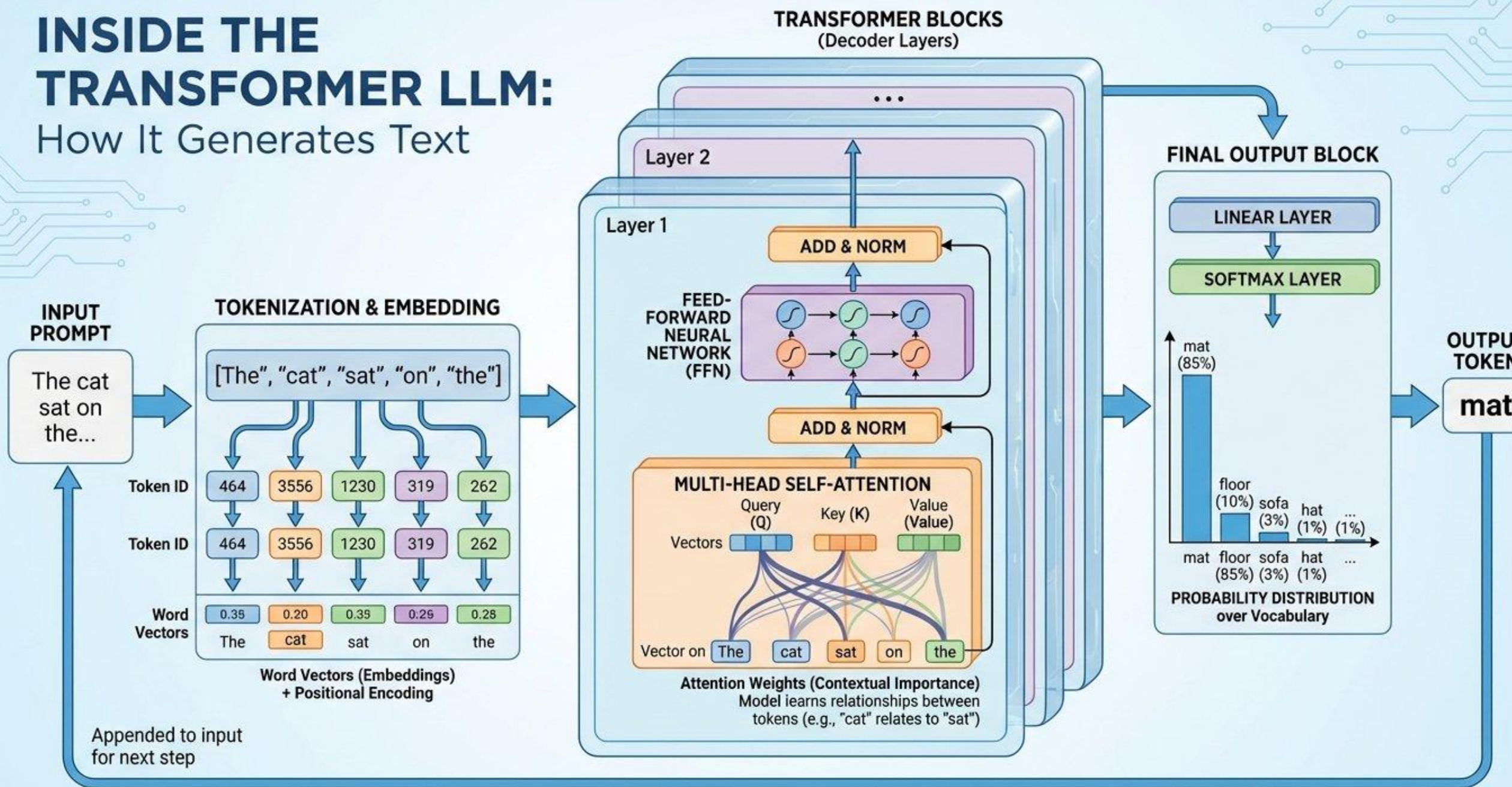
- Rule-based systems, Statistical models, and Recurrent Neural Networks (RNNs).
- *Characteristics:* Task-specific models (one model for translation, a different one for sentiment). Hard to scale.

- **Era 2: Post-Transformer (2017 - Present)**

- The rise of the **Transformer** architecture and **LLMs** (GPT, BERT, Llama).
- *Characteristics:* General-purpose models. One model can translate, summarize, and code.

# INSIDE THE TRANSFORMER LLM:

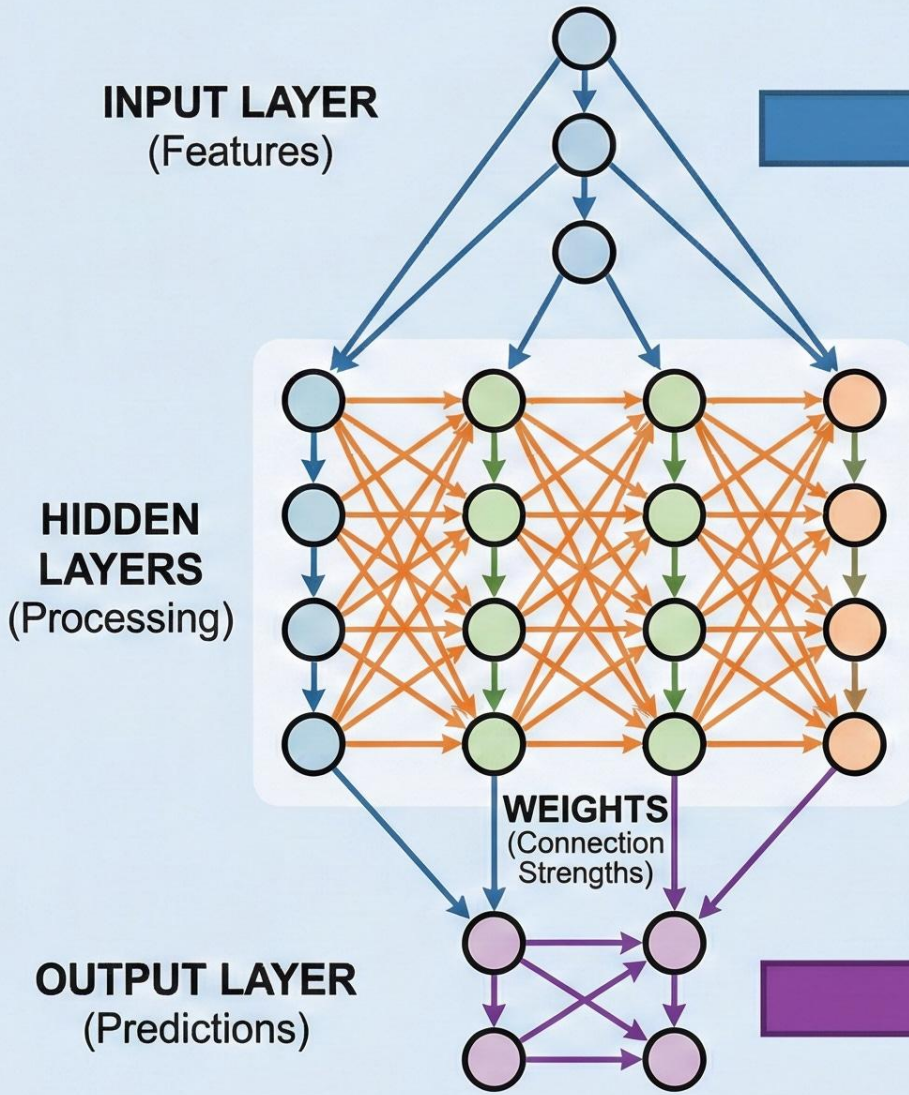
## How It Generates Text



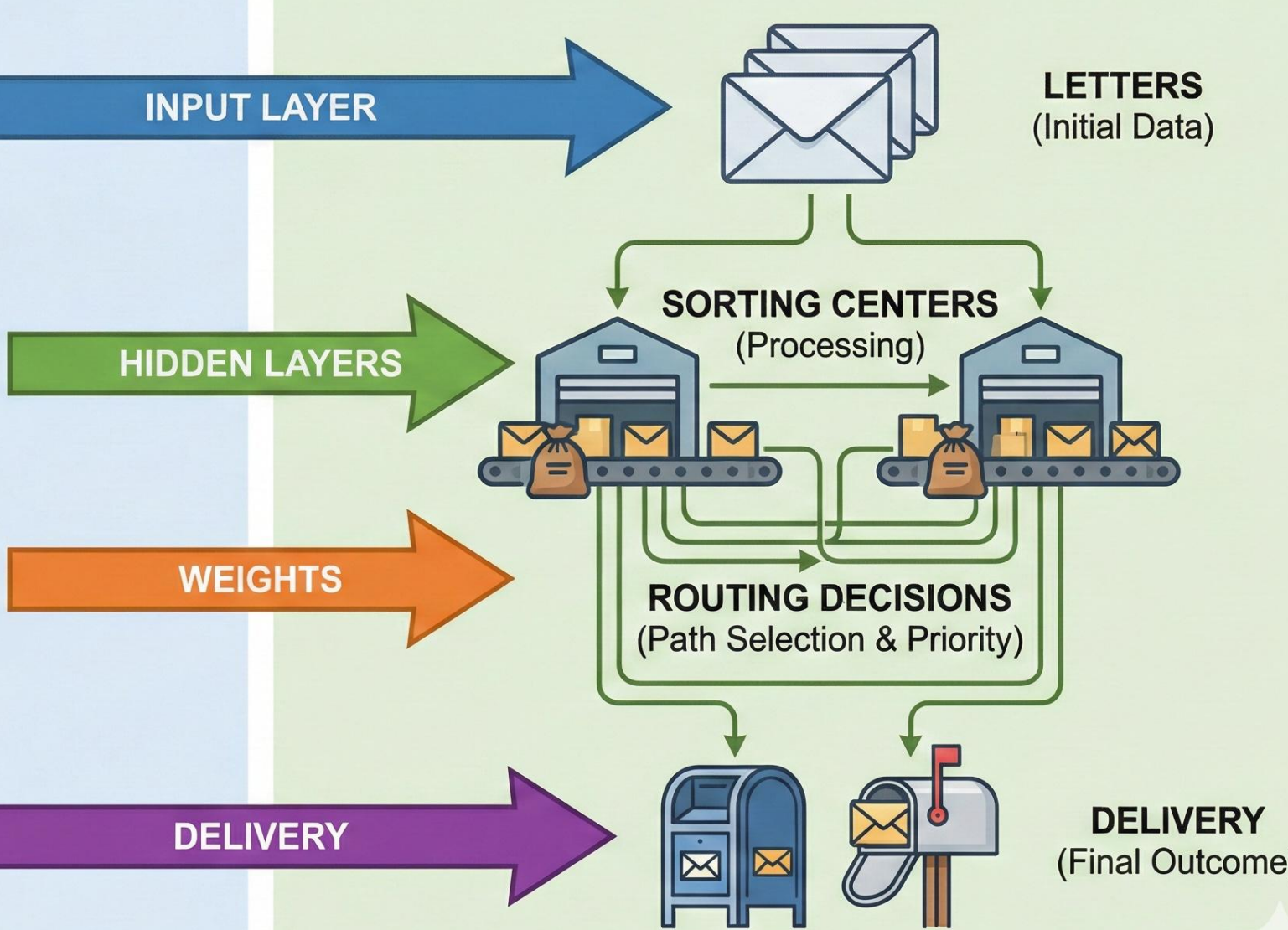


# PARALLEL INFORMATION FLOW: NEURAL NETWORKS & POSTAL SYSTEM

## NEURAL NETWORK STRUCTURE



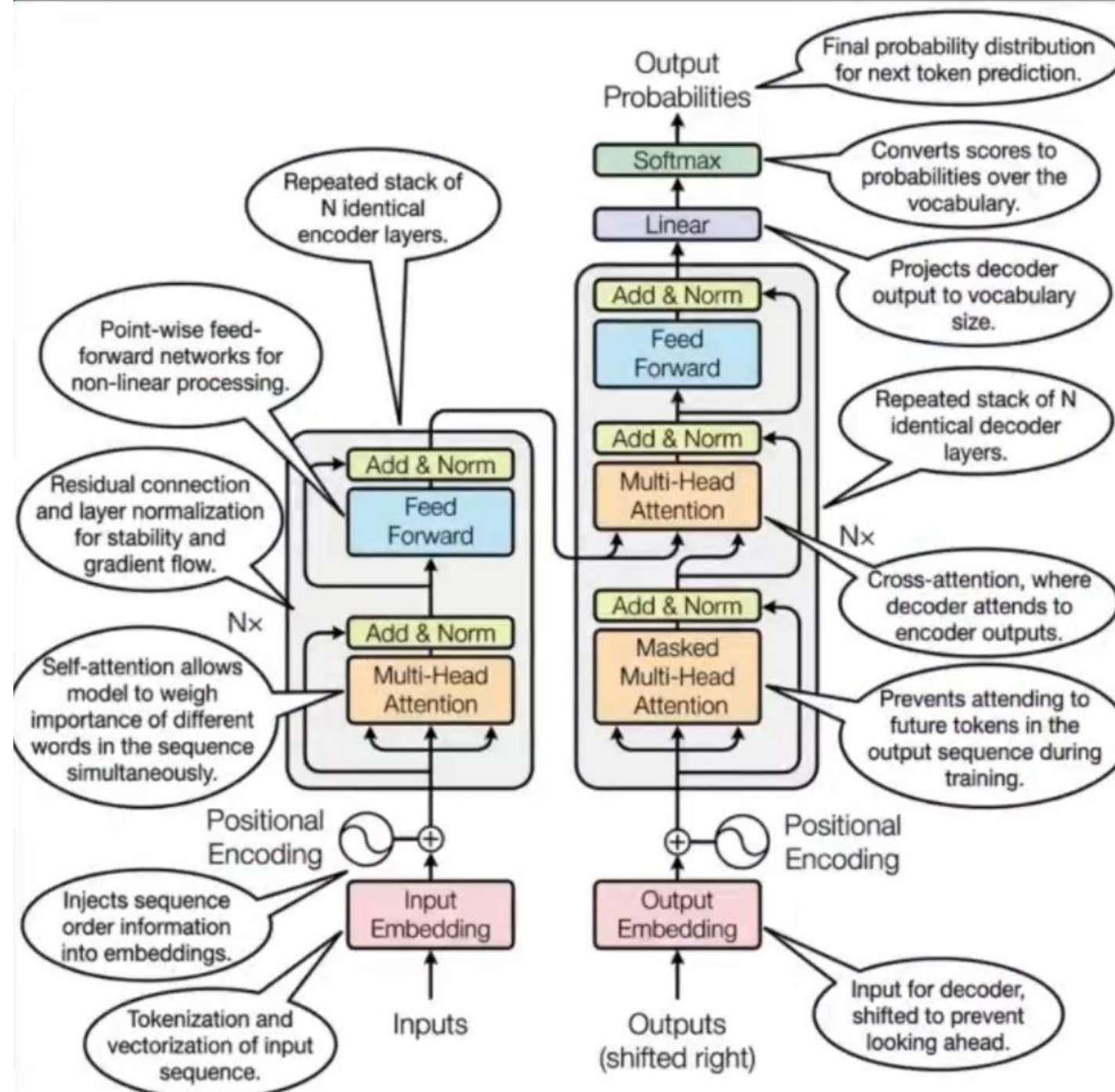
## POSTAL DELIVERY SYSTEM

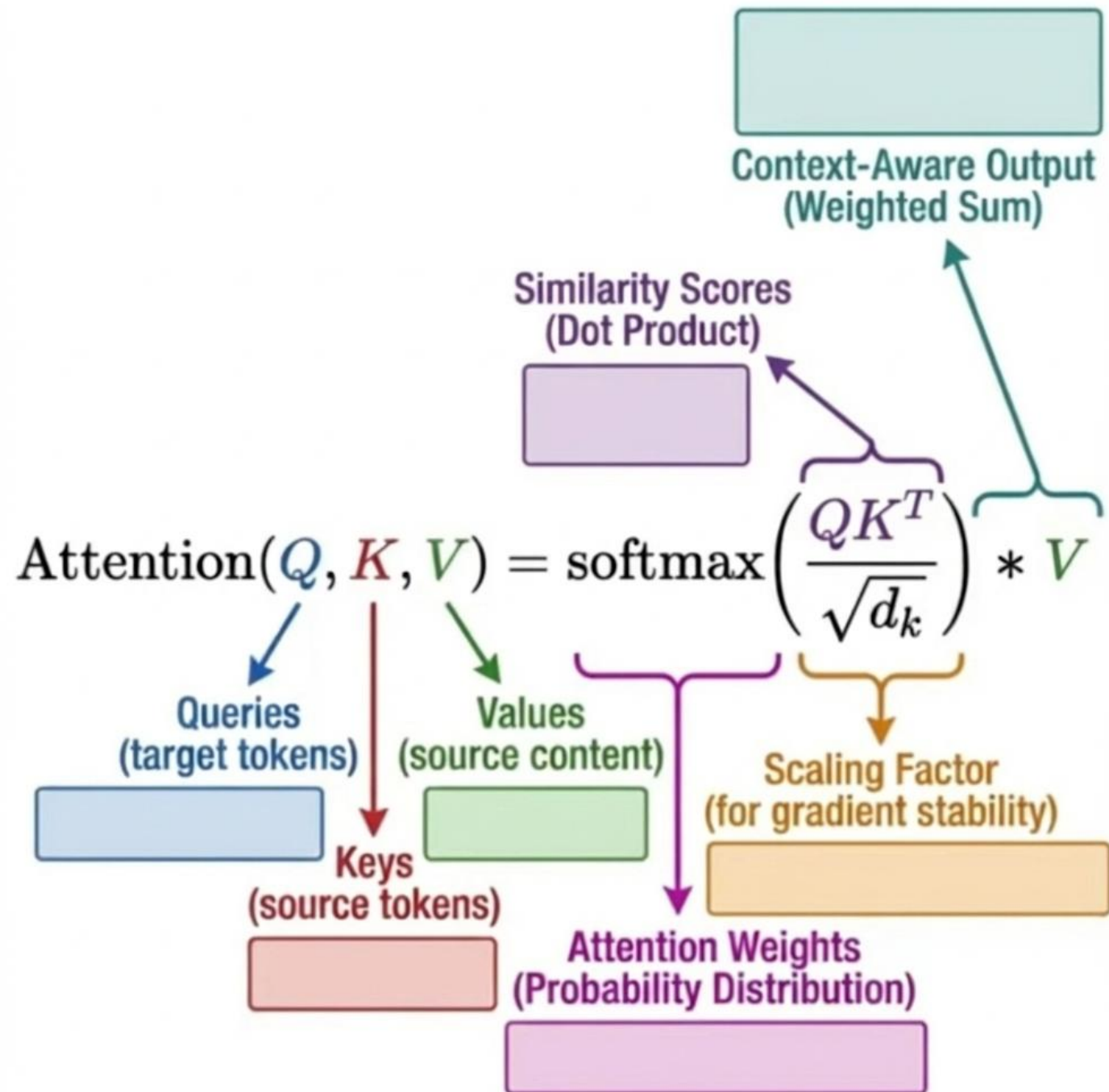


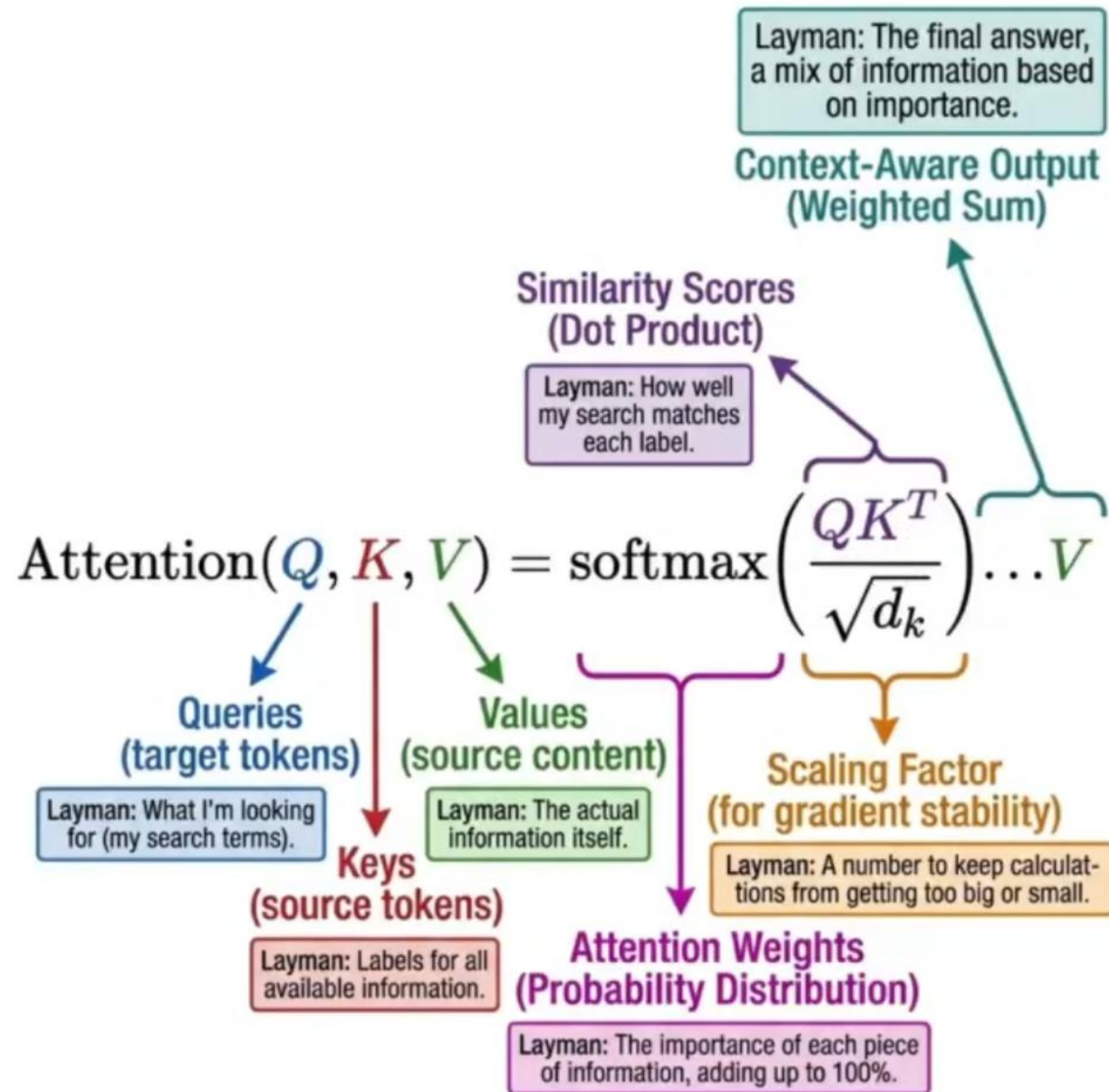
Data flows through layers, transformed by weighted connections.

Mail travels through centers, directed by routing rules.



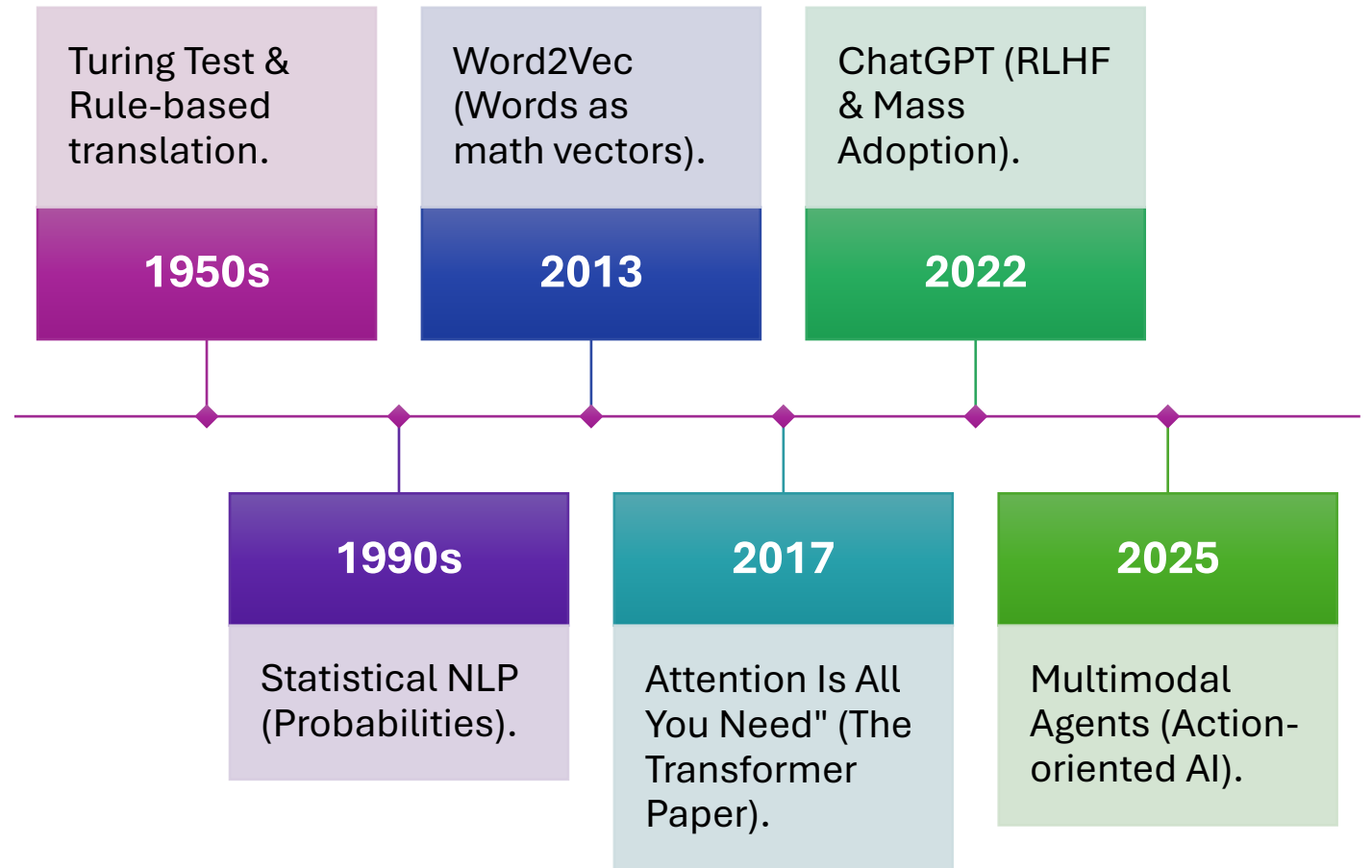








# The Evolution Timeline



# The Paradigm Shift: Computing

## Traditional Programming:

- Input + Explicit Rules (Code) = Output.

## Traditional Machine Learning:

- Input + Output Data = Learned Rules.

## The LLM Paradigm (Prompting):

- Natural Language Instruction + Context = Reasoning & Output.
- *Takeaway*: English is becoming the hottest new programming language.

# Software behind Hardware

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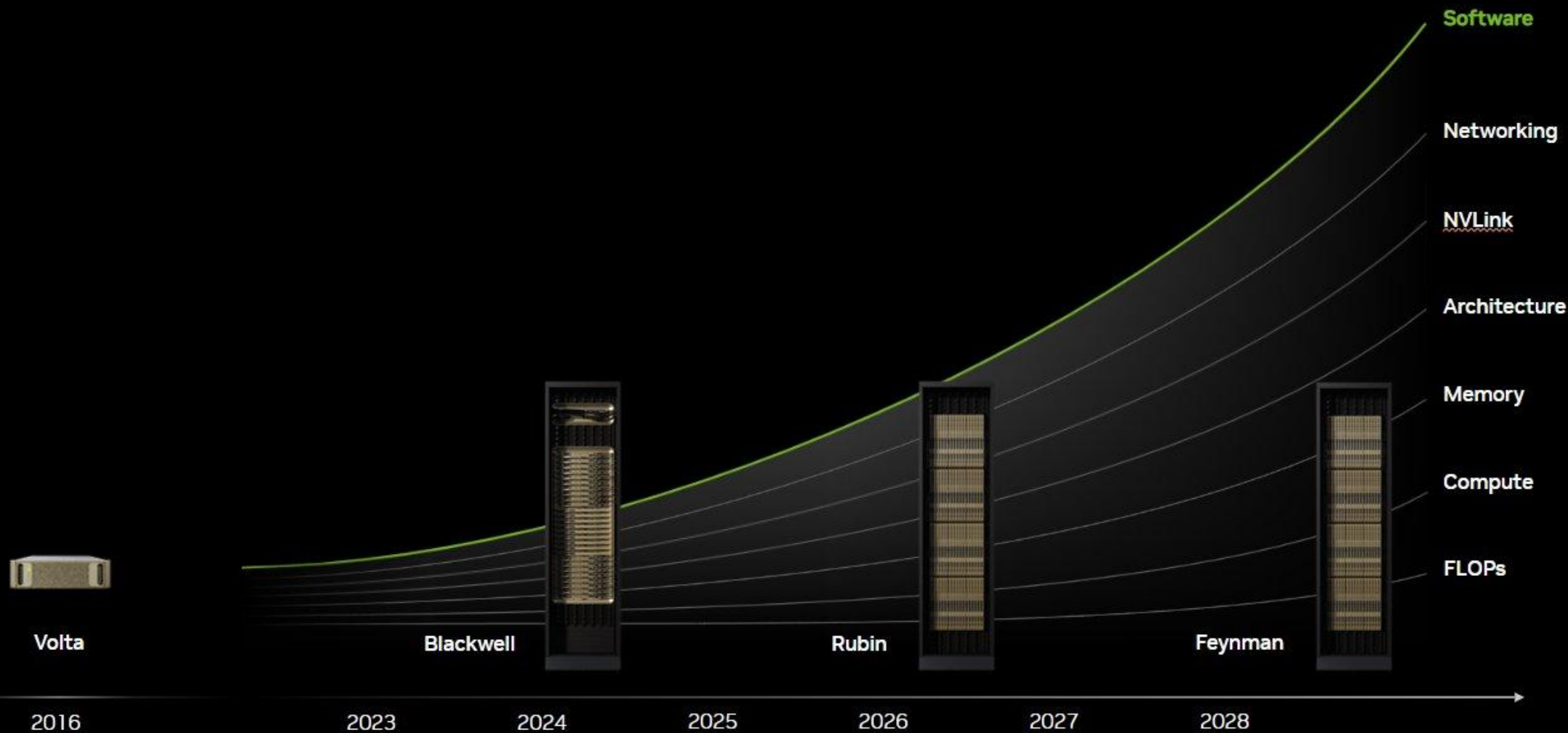




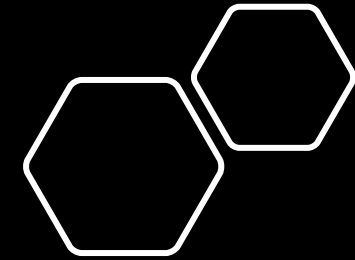
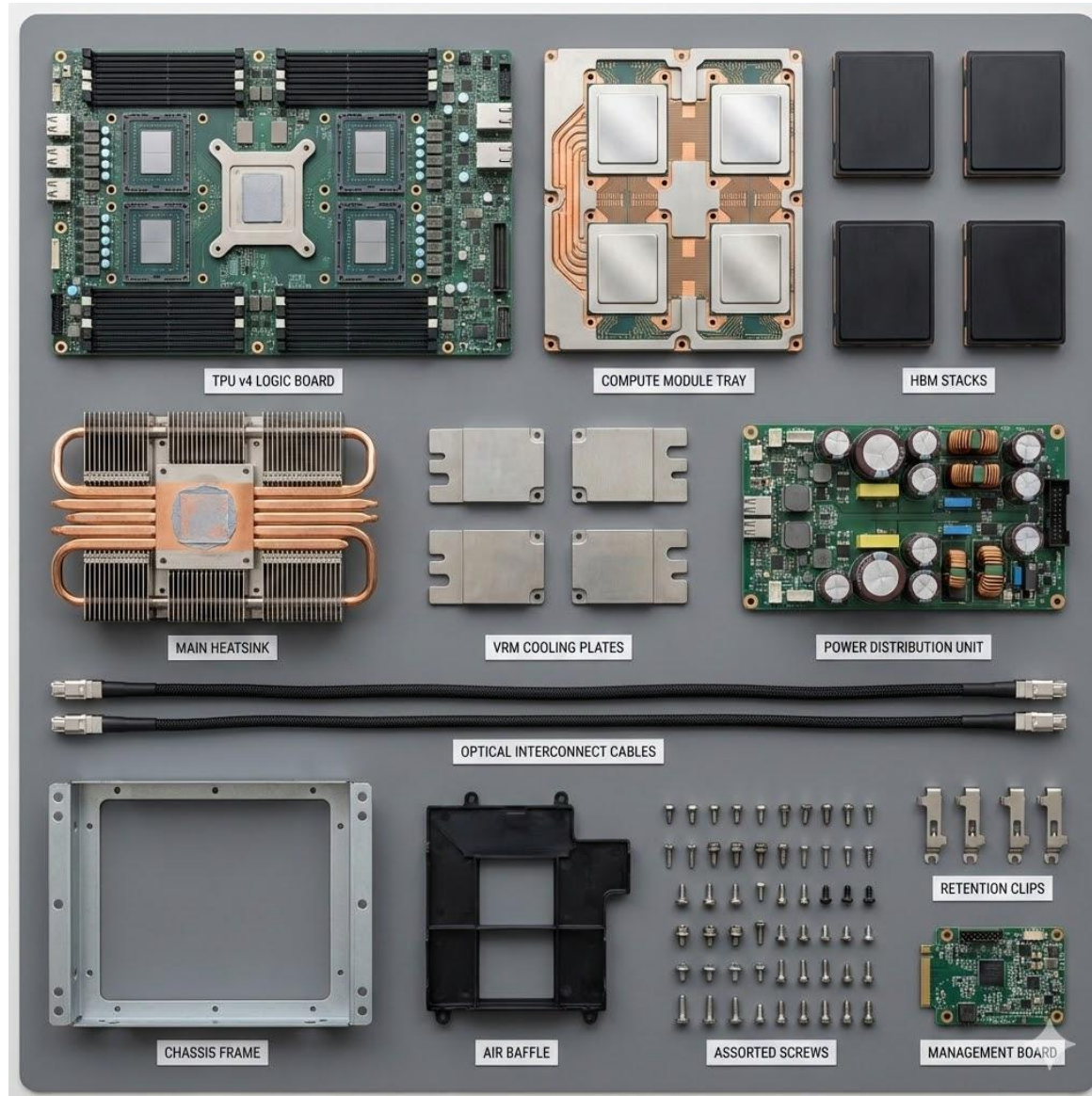
<https://resources.nvidia.com/en-us-blackwell-architecture/blackwell-ultra-datasheet>

# NVIDIA Platform Delivers Continuous Innovation With One-Year Rhythm

Full stack, one architecture, CUDA everywhere



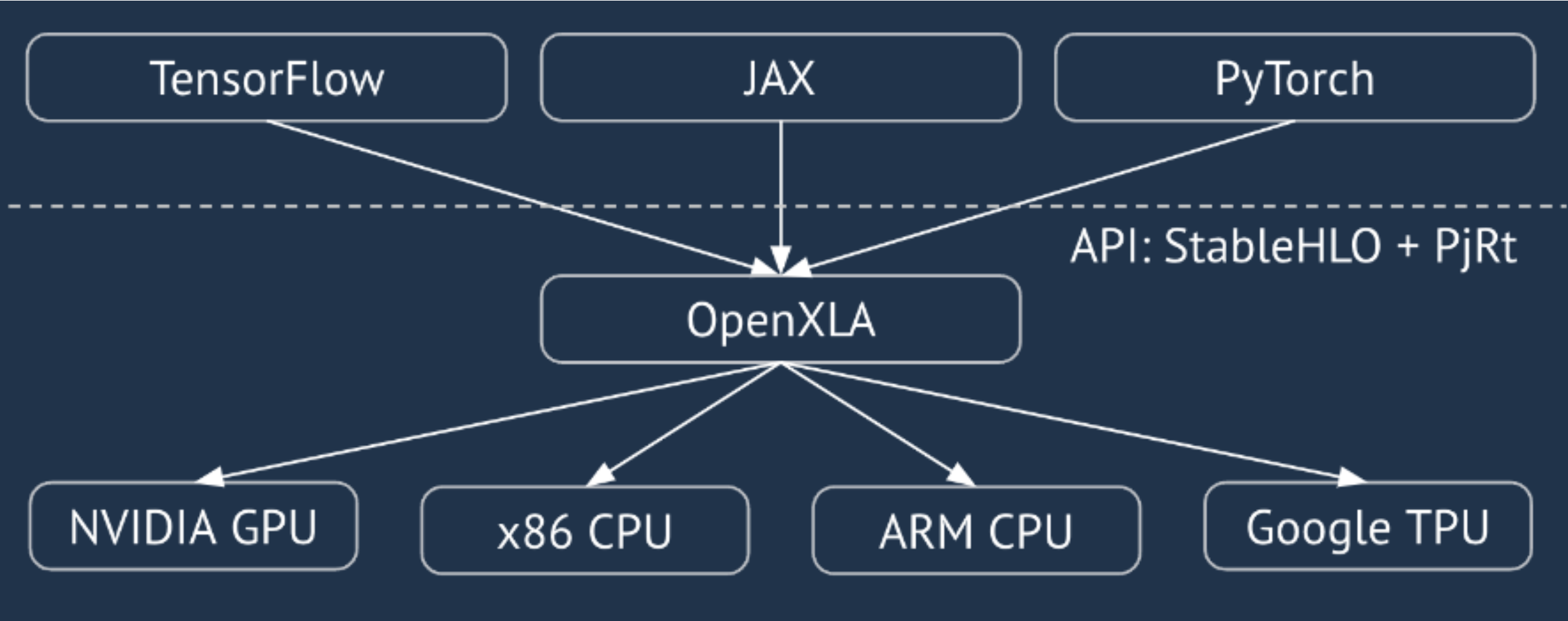




# TPU Knolling by Nano Banana

<https://cloud.google.com/tpu>

# XLA - Accelerated Linear Algebra





# Deconstructing the Pipelines

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# The Traditional NLP Pipeline (The "Old" Way)

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*We still use this for specific, low-cost tasks.*

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**Step 1: Preprocessing** (Cleaning, Tokenizing).

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**Step 2: Feature Extraction** (Counting words, looking for patterns).

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**Step 3: Model** (Logistic Regression, SVM).

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**Step 4: Output** (Class Label).

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*Pros:* Fast, cheap, transparent.

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*Cons:* Can't write poetry or understand deep nuance.

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<https://ai.studio/apps/drive/1xyTLooNlIVa5zHkl6id3yXmVLb9zuBMP>

# The Modern NLP Pipeline (The "New" Way)

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**Step 1: Prompt Engineering** (Designing the instruction).

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**Step 2: Context Injection** (Adding relevant data/documents).

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**Step 3: Black Box Inference** (The LLM does the heavy lifting).

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**Step 4: Output Parsing** (Formatting the text back into data).

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*Pros:* Incredible capabilities, handles unseen tasks.

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*Cons:* Expensive, "Black Box" (hard to explain why), Hallucinations.

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# Anatomy of a Modern App (The RAG Architecture)

- *How do we combine the two?*
- **Scenario:** A Chatbot that answers questions about our University Guidelines.
- **Visual Diagram:**
  - **User Question** ->
  - **Search** (Using Embeddings/Vectors) ->
  - **Retrieve** relevant pages ->
  - **Send Question + Pages** to LLM ->
  - **Answer.**
- **Note:** To build this, you need to understand: **Tokenization** and **Embeddings** (Future Modules).

# Why Learn the Fundamentals?

- If LLMs are so good, why learn Tokenization or Syntax?
- **1. Cost & Latency:** You don't need GPT-4 to classify spam. A simple model is 1000x cheaper and faster.
- **2. Context Window:** LLMs have limits. You need efficient ways (cleaning/preprocessing) to feed them data.
- **3. Debugging:** When the LLM fails, you need to look under the hood.
- **4. Fine-Tuning:** To customize an LLM, you need to understand training loops and data preparation.

# Tools & Lab Intro

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# The Tool stack for this Course

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**Python:** The language of AI.

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**Hugging Face (Transformers):** The "GitHub of AI" – our source for models and datasets.

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**PyTorch:** The deep learning framework.

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**SpaCy / NLTK:** The Swiss-army knives for text processing.

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**Agent Framework (MSFT)/ LangChain / Openai-agents /  
Pydantic-ai/ Google-ADK/ Build or Pick your own:**  
For building chains and agents.



# Ethics First

- **Great Power = Great Responsibility.**
- **Bias:** Models learn from the internet. The internet is biased.
- **Hallucination:** Models can lie confidently.
- **Safety:** We will discuss how to prevent models from generating harmful content.



# Lab Preview: Today's Objective

- **Goal 1: The "Wow" Factor.**
  - Download a Transformer model from Hugging Face.
  - Perform Sentiment Analysis and Translation in 3 lines of code.
- **Goal 2: The Mechanics.**
  - Use NLTK to break a sentence into "Tokens" (words).
  - Compare how an LLM sees text vs. how a human sees text.

# Lab Setup

- **Environment:** Google Colab (Cloud) or Local Jupyter Notebook.
- **Account:** Sign up for Hugging Face (it's free).
- **Library Installation:**
  - pip install transformers
  - pip install nltk
  - pip install spacy

# Example: The Hugging Face "Pipeline"

```
from transformers import pipeline

classifier = pipeline("sentiment-analysis")
result = classifier("I love studying AI and NLP in 2025!")
print(result)

# Output: [{'label': 'POSITIVE', 'score': 0.99}]
```



# Before Next Week



**Read:** "The Illustrated Transformer"  
([Blog post link](#))



**Watch:** [Transformer Explained with Historical reference \(2025\)](#)



**Topic:** We dive into **Text Preprocessing** – Cleaning the data before the AI sees it.



**Q&A**

## **Business Use cases:**

Real world ML and LLM Systems: <https://www.evidentlyai.com/ml-system-design>

## **Microsoft:**

- <https://github.com/microsoft/generative-ai-for-beginners>

## **GitHub:**

- <https://github.com/keon/awesome-nlp>
- <https://github.com/awesomelistsio/awesome-nlp>

## **Google:**

- <https://developers.google.com/machine-learning/crash-course/embeddings>